Whitehead Field Dynamics and Reconfiguration Graphs of Cubic Multigraphs

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Abstract

We study scalar averaging while a connected cubic multigraph changes by Whitehead moves. When an edge $\{i, j\}$ is selected, its endpoint values are both replaced by $(f_i + f_j)/2$. The field variance

$$\mathcal{V}(f) = \sum_i (f_i - \bar{f})^2$$

then decreases by exactly $(f_i - f_j)^2/2$. Identifying an unlabelled target with a chosen representative only permutes the field coordinates and does not change this identity.

For the local-first walk, called Rule A, every one of the $3N$ edge-resolution slots is equally likely. We prove that its stationary distribution is the connected cubic configuration model projected to unlabelled graph classes. For every adapted sequence of connected carriers, the logarithm of the largest possible nonconstant-field norm, divided by the number of steps, is at most $-\pi^2/(6N^3) + O(N^{-5})$. Under the Rule-A stationary law, the random-cubic expansion theorem improves this bound to $-(3-2\sqrt{2}-o(1))/(6N)$. These are field-contraction statements, not mixing estimates for the graph-valued walk. For that base walk, the cut between bridged and bridgeless classes has conductance $2/(3N)(1 + o(1))$, so its relaxation and total-variation mixing times are at least linear in N.

For Rule B, each distinct nonself quotient neighbour receives probability $1/(3N)$ and the unused probability is assigned to holding. This fixed clock is symmetric, has the uniform class law as its stationary distribution, and its operator $I - P_B$ equals the simple-support Laplacian divided by $3N$. It avoids normalizing by the uncontrolled number of neighbours. A Rule-B field process additionally requires a choice of concrete move above each quotient transition. Once all such moves receive positive probability, the fields with zero one-step variance loss are exactly those that are constant on each component of the carrier formed by active edges. It remains open to prove a positive lower bound for $\lambda_2(L_{\mathrm{supp}})$ and to determine the long-time contraction of any specified Rule-B field lift.

On the persistent labelled process, exact finite-time consensus occurs at $N = 2, 4, 8$. Complete unlabelled enumeration through $N = 14$ checks the finite counts, spectra, bridge cut, and Rule-B zero-contraction classes without promoting those finite observations to asymptotic theorems.

1 Introduction and contributions

Fix an even integer N. Let $\mathcal{C}_N$ be the finite set of connected cubic multigraphs on N vertices, considered up to isomorphism unless labels are explicitly introduced. Loops and multiple edges are allowed; a loop contributes two to the degree. In addition to the carrier graph itself, there are three graph-of-graphs conventions on this material. Li's labelled simple graph has fixed-labelled cubic graphs as vertices and collapses repeated moves to the same labelled target. Grave de Peralta–Kolpakov's directed multigraph has unlabelled classes $[G]$ as vertices and retains one directed slot for each unlabelled Whitehead resolution, including multiplicity and self-slots. Finally, the unlabelled

simple-support graph has the same class vertices but collapses all positive nonself multiplicities to one undirected edge. We use Γ_N only when the accompanying text specifies the edge convention; every degree, Laplacian, stationary law, and ball is indexed by its underlying object.

The four state-and-clock conventions used below are summarized as follows.

object	states	one-step data	natural law
carrier field	(G, f)	one carrier edge and resolution	Rule A or specified Rule B
labelled simple GoG	G on $[N]$	distinct labelled targets	degree-proportional
GDPK multiplicity chain	$[G]$	$3N$ resolution slots	$\pi_A \propto c(G)/\lvert\operatorname{Aut}_v(G)\rvert$
unlabelled simple support	$[G]$	one tick per distinct target	fixed $3N$ clock; uniform law.

The number of vertices of this GoG is not an unknown experimental count. If $N = 2n$, then

$$|V(\Gamma_N)| = a(n),$$

where $a(n)$ is OEIS A005967, the number of isomorphism classes of connected 3-regular multigraphs of order $2n$, loops allowed [3]. The first values are

$$2, 5, 17, 71, 388, 2592, 21096, \ldots$$

for $N = 2, 4, 6, 8, 10, 12, 14, \ldots$. Thus enumeration code has a hard external sanity check before any field dynamics is studied.

The papers of Grave de Peralta–Kolpakov and Li establish substantial large-scale structure. Grave de Peralta–Kolpakov prove that the outer-conductance of Γ_N tends to zero, using the bottleneck given by bridged cubic graphs [1]. Li shows that the graph of Whitehead moves has increasingly many bounded Laplacian eigenvalues, using many local cycle bottlenecks [2]. These results concern ordinary functions on the vertex set of Γ_N.

Here the moving object is different. Over a cubic graph $G = (V, E)$ we place a scalar field

$$f : V \to \mathbb{R}.$$

So the fibre over the class $\gamma = [G]$ is

$$\mathcal{F}_\gamma = C^0(G) \simeq \mathbb{R}^{V(G)}.$$

When a Whitehead move is performed on an edge $e = \{i, j\}$, the graph changes and the field is updated by

$$f_i, f_j \mapsto \frac{f_i + f_j}{2}, \frac{f_i + f_j}{2}.$$

Afterwards one may resettle the new graph onto a chosen representative of its isomorphism class; this only permutes field coordinates.

The field is scalar because no differential data is attached to the vertices of G; the vector space $\mathbb{R}^N$ is only its coordinate representation.

Main results. The conclusions are grouped by the state space and by the random rule.

1. For the carrier field, we prove the exact variance loss at each averaged edge. We show how to pull every settlement permutation back to a persistent coordinate frame. Connectedness alone bounds the top field exponent by $-\pi^2/(6N^3) + O(N^{-5})$. Typical expansion under stationary Rule A improves the bound to $-(3 - 2\sqrt{2} - o(1))/(6N)$. On the persistent labelled process, explicit averaging words give exact finite-time consensus at $N = 2, 4, 8$.

2. For Li's fixed-labelled simple graph, we keep separate the walk that chooses a distinct neighbour, the walk with a uniform $6N$-slot clock, and the walk that retains move multiplicity. We prove that at least $N(N-3)/2$ nonconstant eigenvalues of the combinatorial Laplacian stay bounded, compute fixed-radius ball growth, and obtain linear lower bounds for diameter and typical distance.

3. For the Rule-A chain on unlabelled classes, we prove the exact configuration-model stationary law and show that its spectrum lies in $[-1/2, 1]$. We compute the Dirichlet form of loop count. We prove that the bridged-class cut has conductance $2/(3N)(1 + o(1))$, which gives an $\Omega(N)$ relaxation time and an explicit $\Omega(N)$ lower bound for total-variation mixing.

4. For Rule B on the unlabelled simple support, we assign probability $1/(3N)$ to every distinct nonself neighbour and assign the remaining probability to holding. The resulting kernel is symmetric, has uniform stationary law, and satisfies $I - P_B = L_{\mathrm{supp}}/(3N)$. This removes the need to estimate the variable support degrees. The degree-normalized walk remains available as the jump chain conditioned on moving. For any specified full-support field lift, we identify all fields with zero one-step variance loss by the connected components of the active carrier edges.

The theoretical statements are separated below from complete finite enumeration, conjectures, and the limited Lean verification layer.

2 Whitehead Moves and the Exact Random Rule

Let $e = \{u, v\}$ be a non-loop edge of a cubic graph. At u there are exactly two other incident half-edges, call them a, b, and at v there are exactly two other incident half-edges, call them c, d. Locally the edge e sees the partition

$$\{a, b\} \mid \{c, d\}.$$

A Whitehead move collapses e to a four-valent vertex and re-expands in one of the two other pairings:

$$\{a, c\} \mid \{b, d\}, \qquad \{a, d\} \mid \{b, c\}.$$

On a fixed labelled vertex set, each of these two unordered pairings has two assignments to the labelled central vertices u, v. Thus there are four labelled moves per non-loop edge, as in Li's labelled graph, but only two unlabelled resolution slots per edge, as in Grave de Peralta–Kolpakov. For a cubic graph on N vertices the corresponding raw counts are $6N$ and $3N$. If e is a loop, we use the convention of [1]: the Whitehead move is trivial. In the half-edge model this convention is natural because the pairing of half-edges is fixed and only the assignment of nearby half-edges to vertices changes; when the two endpoints coincide there is no reassignment.

For the averaged field the factor of two can be compressed. The two labelled orientations of one unlabelled resolution differ by the transposition $\tau = (uv)$, and

$$P_\tau A_{uv} = A_{uv}.$$

With compatible settlements they therefore produce the same transported field. At a target with automorphisms they agree modulo a target automorphism. The transition table may consequently store two resolution records per edge without losing the base-class kernel or the one-step variance/source-Laplacian calculation. For a full trajectory or a curvature calculation, however, an intermediate target automorphism can change later averages. The compression is lossless there only after specifying coherent settlements (or explicitly quotienting the fibre by stabilizers); otherwise

the omitted stabilizer holonomy is dynamical data. The compression also does not identify the two targets in the labelled simple graph and must not be used for labelled ball counts.

There are two inequivalent random rules.

Definition 1 (Rule A: local move first)**.** Start from a settled representative of the unlabelled class $\gamma = [G]$. Choose a labelled Whitehead move upstairs, or equivalently for this averaged field choose one of the two compressed unlabelled resolutions of an edge object. Apply this move to get a labelled graph H, project to the unlabelled class $\delta = [H]$, and choose an isomorphism settling H onto the chosen representative of δ.

Rule A weights unlabelled transitions by concrete multiplicity. If many different edges and resolutions lead from γ to the same class δ, then that edge of the unlabelled GoG is traversed more often. This is the natural rule if the physics is local in the current cubic graph.

Definition 2 (Rule B: fixed support clock)**.** Start at an unlabelled class G. For every distinct nonself class H that can be reached by at least one Rule-A slot, assign one of the $3N$ clock ticks to the transition $G \to H$. Assign the remaining $3N - q(G)$ ticks to holding at G, where

$$q(G) = \#\{H \neq G : m_{GH} > 0\}.$$

Thus

$$P_B(G,H) = \frac{1}{3N} \quad (H \neq G,\ m_{GH} > 0), \qquad P_B(G,G) = 1 - \frac{q(G)}{3N}.$$

The support is undirected, so P_B is symmetric and the uniform law on unlabelled classes is stationary. Moreover

$$I - P_B = \frac{1}{3N} L_{\mathrm{supp}}.$$

This fixed clock removes the variable-degree normalization from the Rule-B spectrum. The walk that chooses one of the $q(G)$ neighbours uniformly is the jump chain obtained by conditioning Rule B on leaving G:

$$J_B(G,H) = \frac{1}{q(G)} \quad (H \neq G,\ m_{GH} > 0).$$

It has stationary mass proportional to $q(G)$. We keep J_B as a named comparison chain, but it is not the default Rule-B clock.

Rule B still requires a second choice before it transports a field. A tick labelled by $G \to H$ specifies the target class but not which concrete edge and resolution above that transition is used. A holding tick leaves both carrier and field unchanged. Under a non-holding Rule-B tick, the edge of G being averaged need not be uniform among the edge objects of G. Its expected field step is therefore governed by a weighted Laplacian rather than the ordinary Laplacian of G.

This distinction controls both the base transition law and the field operator. Rule A is local first: every formal edge-resolution slot in the current cubic graph is equally likely, so every edge object is equally likely to be averaged. Rule B retains only one tick for each distinct target. If ten Rule-A slots land in one quotient neighbour and one slot lands in another, Rule B gives each target one tick and converts the other nine ticks into holding. A lift rule then distributes a target's single tick among its concrete realizations.

For a fixed class $\gamma = [G]$, the Rule B field step is therefore recorded as a weighted source Laplacian

$$L_\gamma^B = \sum_{e=\{i,j\}} p_\gamma^B(e)(e_i - e_j)(e_i - e_j)^T,$$

where $p_\gamma^B(e)$ is the total Rule B probability that the chosen lift averages edge e. Rule A corresponds to the special case

$$p_\gamma^A(e) = \frac{1}{|E(G)|} = \frac{2}{3N}$$

for every edge object. Numerically, the right question is not simply whether the base chains on Γ_N differ, but whether L_γ^B differs substantially from $(2/3N)L_G$ on the states that the chain actually visits.

Remark 3 (What labels are doing)**.** The base graph Γ_N can stay unlabelled. Labels on the vertices of the current cubic graph are accounting coordinates for the field. A step consists of choosing a labelled lift, applying a concrete move, and then resettling onto some representative of the target isomorphism class. When automorphisms are trivial, the settlement ambiguity essentially disappears once representatives are fixed. When automorphisms are nontrivial, the ambiguity is an orthogonal permutation of the fibre, hence a gauge ambiguity rather than an extra contraction mechanism.

Proposition 4 (Configuration-model stationary law of Rule A)**.** For a labelled adjacency multigraph G, let m_{ii} be the number of loops at i, let m_{ij} be the number of edges between $i < j$, and put

$$c(G) = \frac{(3!)^N}{\prod_i 2^{m_{ii}} m_{ii}! \prod_{i<j} m_{ij}!}.$$

Let $\mathrm{Aut}_v(G)$ denote the vertex-permutation automorphism group. If m_{GH} is the number of the $3N$ compressed Grave de Peralta–Kolpakov slots from a representative of $[G]$ to the class $[H]$, then

$$\frac{c(G)}{|\mathrm{Aut}_v(G)|} m_{GH} = \frac{c(H)}{|\mathrm{Aut}_v(H)|} m_{HG}.$$

Consequently the Rule-A class chain

$$P_A([G],[H]) = \frac{m_{GH}}{3N}$$

is reversible with stationary law

$$\pi_A([G]) = \frac{c(G)/|\mathrm{Aut}_v(G)|}{\sum_{[K]\in\Gamma_N} c(K)/|\mathrm{Aut}_v(K)|}.$$

This is exactly the cubic configuration-model law, conditioned on connectedness and projected to unlabelled classes.

Proof. Give each labelled vertex three distinguishable half-edges. The number of perfect matchings of these half-edges which realize a fixed labelled adjacency multigraph G is $c(G)$: the denominator removes the two orientations of each loop, permutations of loops at a vertex, and permutations of parallel edges. An unlabelled class has $N!/|\mathrm{Aut}_v(G)|$ labelled adjacency realizations.

On fully labelled half-edge configurations, retain the four endpoint-oriented moves at a non-loop edge (and four duplicate holding slots at a loop). Each move has a unique reverse, so uniform measure on configurations is reversible. The four oriented moves collapse in pairs to the two unlabelled resolutions recorded by m_{GH}. Lumping the reversible configuration chain by isomorphism class gives the displayed reciprocity identity and stationary law. Whitehead moves preserve connectedness, so restricting and conditioning on the connected component is legitimate. □

Remark 5. Thus stationarity is not a free assumption, and there is no third unknown Rule-A graph distribution: its carrier marginal is the configuration model. The weights $c(G)$ are essential when loops and parallel edges occur. Already for $N = 2$, the triple-edge class and the class with one joining edge and one loop at each endpoint have equal vertex-automorphism orders but stationary masses 2/5 and 3/5.

3 Projection Calculus

For a labelled vertex set $V = \{1, \dots, N\}$ and a non-loop edge $e = \{i, j\}$, define the local averaging operator

$$(A_{ij}f)_i = (A_{ij}f)_j = \frac{f_i + f_j}{2}, \qquad (A_{ij}f)_k = f_k \quad (k \neq i, j).$$

Let e_i be the ith coordinate vector and put

$$u_{ij} = \frac{e_i - e_j}{\sqrt{2}}.$$

Then

$$A_{ij} = I - u_{ij}u_{ij}^T = I - \frac{1}{2}(e_i - e_j)(e_i - e_j)^T.$$

Lemma 6 (Both pieces are projections)**.** A_{ij} is the orthogonal projection onto the hyperplane

$$H_{ij} = \{f : f_i = f_j\}.$$

The complementary operator

$$I - A_{ij} = \frac{1}{2}(e_i - e_j)(e_i - e_j)^T$$

is the orthogonal projection onto the line $\mathbb{R}(e_i - e_j)$.

Proof. The vector u_{ij} is a unit normal to H_{ij}. Therefore $u_{ij}u_{ij}^T$ is the orthogonal projection onto $H_{ij}^{\perp}$ and $I - u_{ij}u_{ij}^T$ is the orthogonal projection onto H_{ij}. This also gives $A_{ij}^2 = A_{ij}$ and $(I - A_{ij})^2 = I - A_{ij}$. □

Thus neither A_{ij} nor $I - A_{ij}$ is nilpotent. The contrast removed by the averaging step is

$$(I - A_{ij})f = \frac{f_i - f_j}{2}(e_i - e_j).$$

This expression does not require an orientation of the edge: exchanging i and j changes both factors by a sign.

The basic noncommutativity is already visible on three vertices. On $\operatorname{span}\{e_1, e_2, e_3\}$,

$$[A_{23}, A_{12}] = A_{23}A_{12} - A_{12}A_{23} = \frac{1}{4}\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Consequently

$$\operatorname{Spec}_{\text{sing}}([A_{23}, A_{12}]) = \left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}}{4}, 0\right\}.$$

This is the local algebraic seed of curvature: disjoint averages commute, incident averages generally do not.

Lemma 7 (Commutation criterion for elementary averages)**.** The projections A_{ij} and $A_{k\ell}$ commute if the unordered pairs $\{i,j\}$ and $\{k,\ell\}$ are equal or disjoint. If they share exactly one vertex, they do not commute.

Proof. If the pairs are equal this is trivial. If they are disjoint, the operators act on disjoint coordinate blocks. For incident pairs it suffices, after relabeling, to compute the commutator above, which is nonzero. □

4 Variance and Heat Step

Define the mean and variance by

$$\bar{f} = \frac{1}{N}\sum_{k=1}^{N} f_k, \qquad \mathcal{V}(f) = \sum_{k=1}^{N}(f_k - \bar{f})^2 = \left\|f - \bar{f}\mathbf{1}\right\|^2.$$

The constants are the consensus subspace. The variance is the squared distance from this subspace.

Lemma 8 (One-step Lyapunov identity)**.** For one averaging step on $\{i,j\}$,

$$\sum_{k=1}^{N}(A_{ij}f)_k = \sum_{k=1}^{N} f_k$$

and

$$\mathcal{V}(A_{ij}f) = \mathcal{V}(f) - \frac{1}{2}(f_i - f_j)^2.$$

Proof. The sum of the i,j coordinates is preserved. Let $g = f - \bar{f}\mathbf{1}$. Since $A_{ij}\mathbf{1} = \mathbf{1}$, the mean-zero part after averaging is $A_{ij}g$. Since A_{ij} is an orthogonal projection,

$$\|A_{ij}g\|^2 = \|g\|^2 - \langle g, u_{ij}\rangle^2 = \mathcal{V}(f) - \frac{1}{2}(f_i - f_j)^2.$$

□

Let L_G be the combinatorial Laplacian of the current cubic multigraph:

$$L_G = \sum_{\{i,j\}\in E(G),\, i\neq j} (e_i - e_j)(e_i - e_j)^T.$$

Loops contribute zero to this sum. Parallel edges are counted with multiplicity. If an edge object is chosen uniformly from all $|E(G)| = 3N/2$ edge objects, then

$$\mathbb{E}[A_{ij}f \mid G, f] = \left(I - \frac{1}{2|E(G)|}L_G\right) f = \left(I - \frac{1}{3N}L_G\right) f,$$

and

$$\mathbb{E}[\mathcal{V}(f') \mid G, f] = \mathcal{V}(f) - \frac{1}{2|E(G)|} f^T L_G f = \mathcal{V}(f) - \frac{1}{3N} f^T L_G f.$$

If $\lambda_2(L_G) \geq c_0$, the definition of the second eigenvalue gives

$$\mathbb{E}[\mathcal{V}(f') \mid G, f] \leq \left(1 - \frac{c_0}{3N}\right)\mathcal{V}(f).$$

More generally, assign weights $\mu(e) \geq 0$ to edge objects with $\sum_e \mu(e) \leq 1$, and use the remaining probability for a holding step. The same calculation uses the weighted Laplacian

$$L_{G,\mu} = \sum_{\{i,j\} \in E(G),\, i \neq j} \mu(\{i,j\})(e_i - e_j)(e_i - e_j)^T$$

and gives

$$\mathbb{E}[\mathcal{V}(f') \mid G, f] = \mathcal{V}(f) - \frac{1}{2} f^T L_{G,\mu} f.$$

For fixed-clock Rule B, the active weights sum to $q(G)/(3N)$; the other $1 - q(G)/(3N)$ of the probability is holding. Thus Rule B replaces the ordinary carrier Laplacian by a source-dependent weighted Laplacian.

5 The Field Bundle and Gauge

The unlabelled graph-of-graphs has no canonical coordinates in its fibres. The right object is the bundle

$$\gamma = [G] \in \Gamma_N, \qquad \mathcal{F}_\gamma = C^0(G).$$

Choosing a representative $C(\gamma)$ for every class and choosing settlement isomorphisms after every move is a choice of gauge.

For a concrete Whitehead move $m : \gamma \to \delta$, a settlement gives a linear map

$$T_m : \mathcal{F}_\gamma \to \mathcal{F}_\delta, \qquad T_m = P_\sigma A_{ij},$$

where P_σ is the permutation matrix induced by the settling isomorphism. If the source and target frames are changed by permutation matrices Q_γ, Q_δ, then the same transport is represented by

$$T_m \mapsto Q_\delta T_m Q_\gamma^{-1}.$$

Thus matrix entries of T_m are not intrinsic. Norms, ranks, and singular values of endpoint-covariant expressions are intrinsic.

Lemma 9 (Gauge does not affect the Lyapunov identity)**.** For one settled unlabelled step

$$f' = P_\sigma A_{ij} f,$$

the mean is preserved and

$$\mathcal{V}(f') = \mathcal{V}(f) - \frac{1}{2}(f_i - f_j)^2.$$

Proof. The permutation P_σ preserves the constant vector, coordinate sums, and Euclidean norm. Apply the labelled variance identity before the permutation. □

This is the reason the one-step Lyapunov identity is clean: settlement choices are orthogonal gauge changes, not hidden dissipation. They are nevertheless pathwise dynamical data, because cumulative settlements conjugate the endpoint pairs averaged at later times. The exact pullback formula below separates these two statements.

Proposition 10 (Exact settlement pullback)**.** Let $T_s = P_s A_{e_s}$, where P_s is an arbitrary settlement permutation. Put

$$Q_0 = I, \qquad Q_{s+1} = P_s Q_s, \qquad \widetilde{e}_s = Q_s^{-1} e_s.$$

Then

$$T_{t-1} \cdots T_0 = Q_t A_{\widetilde{e}_{t-1}} \cdots A_{\widetilde{e}_0}.$$

Consequently the singular values of a settled product are exactly those of a product of ordinary pair averages in one persistent frame.

Proof. If $T_{s-1} \cdots T_0 = Q_s Y_s$, then

$$Y_{s+1} = Q_{s+1}^{-1} P_s A_{e_s} Q_s Y_s = Q_s^{-1} A_{e_s} Q_s Y_s = A_{Q_s^{-1} e_s} Y_s.$$

Induction proves the factorization. □

6 Covariant Difference and Square Curvature

Let s be a section of the field bundle over some portion of Γ_N, so $s_\gamma \in \mathcal{F}_\gamma$. Given a concrete move $m : \gamma \to \delta$, define the covariant difference

$$(Ds)_m = s_\delta - T_m s_\gamma.$$

These maps are not invertible: every nonloop T_m has rank $N-1$. They therefore form a dissipative semigroup cocycle rather than parallel transport for a vector-bundle connection in the standard sense. Comparing two paths with the same endpoints is still meaningful, and squares give a useful local comparison, but they are not the first obstruction to path independence.

Consider two concrete two-step paths with common source and target:

$$p : \gamma \to \alpha \to \delta, \qquad q : \gamma \to \beta \to \delta.$$

After choosing local settlements, put

$$T_p = T_{p_2} T_{p_1}, \qquad T_q = T_{q_2} T_{q_1},$$

and define the square defect

$$\Omega(p, q) = T_p - T_q.$$

Proposition 11 (Gauge covariance of the square defect)**.** Under a coherent gauge change at the common source and common target,

$$\Omega(p, q) \mapsto Q_\delta \Omega(p, q) Q_\gamma^{-1}.$$

Consequently

$$\mathrm{Spec}_{\mathrm{sing}}(T_p - T_q), \quad \|T_p - T_q\|_{\mathrm{op}}, \quad \|T_p - T_q\|_{\mathrm{Frob}}, \quad \mathrm{rank}(T_p - T_q)$$

are independent of the chosen source and target frames.

Proof. Both path transports transform by the same endpoint permutations:

$$T_p \mapsto Q_\delta T_p Q_\gamma^{-1}, \qquad T_q \mapsto Q_\delta T_q Q_\gamma^{-1}.$$

Subtract. Left and right multiplication by orthogonal permutation matrices preserves singular values, operator norm, Frobenius norm, and rank. □

Thus the clean curvature definition is not a matrix of entries. It is the singular spectrum of the square defect:

$$\mathrm{Curv}(p,q) = \mathrm{Spec}_{\mathrm{sing}}(T_p - T_q),$$

or, as a scalar,

$$\kappa(p,q) = \|T_p - T_q\|_{\mathrm{op}}.$$

The elementary operators A_{ij} and $I - A_{ij}$ are projections, not differentials. The displayed D is currently defined only from vertices to edges. Calling the square defect the obstruction to $D^2 = 0$ would require a separate extension of D from edges to chosen 2-cells; without that extra definition, the justified statement is gauge covariance of the path difference.

There is already a one-dimensional backtracking defect. If a Whitehead move on e is followed by its graph inverse with coherent inverse settlement, then in the original frame

$$T_{m^{-1}} T_m = A_e^2 = A_e \neq I,$$

and $I - A_e$ is a rank-one orthogonal projection of operator norm one. Thus even a contractible move–inverse path is dissipative.

Proposition 12 (Flatness for disjoint moves)**.** If two concrete Whitehead moves are performed on non-loop edges whose endpoint sets are disjoint, then the corresponding labelled square defect is zero. Any coherently settled unlabelled square has

$$\mathrm{Spec}_{\mathrm{sing}}(T_p - T_q) = \{0, \ldots, 0\}.$$

Proof. In the half-edge model, each Whitehead move only reassigns the four half-edges incident to the endpoints of the selected edge and not belonging to that edge. Disjoint endpoint sets give disjoint local data, so the graph rewrites commute. The field operators also commute because the averaging projections act on disjoint coordinate pairs. Thus both two-step path transports are identical, up to the same final settlement. □

Incident squares are the basic non-flat squares. Their defect is not only a property of the cubic graph rewrite; it also includes the averaging operators. Locally, when the two consecutive averaged pairs share one vertex, the commutator has singular values

$$\frac{\sqrt{3}}{4}, \frac{\sqrt{3}}{4}, 0$$

on the three affected coordinates. In an actual GoG square this local algebra is dressed by the two graph rewrites and the endpoint settlements.

Automorphisms create one extra subtlety. If the endpoint graph has a nontrivial automorphism group, two apparently different final settlements may differ by $U \in \mathrm{Aut}(\delta)$. A quotient-level aligned defect is therefore

$$\kappa_{\mathrm{orb}}(p,q) = \min_{U \in \mathrm{Aut}(\delta)} \|T_p - U T_q\|_{\mathrm{op}}.$$

This is not needed when the two lifted paths and settlements are tracked coherently, but it is useful when comparing defects computed from canonical representatives. Averaging over all settlements,

$$\overline{T}_m = \frac{1}{|\mathrm{Aut}(\delta)|} \sum_{\alpha \in \mathrm{Aut}(\delta)} P_\alpha T_m,$$

is canonical but changes the dynamics: it projects toward automorphism-invariant components and can introduce artificial contraction.

7 Deterministic Consensus from Products of Projections

Before considering random graph motion, isolate the deterministic fact about averaging projections. Let

$$E_0 = \mathbf{1}^\perp = \{f \in \mathbb{R}^N : \sum_i f_i = 0\}.$$

Every A_{ij} preserves E_0 and has norm at most one on E_0.

Lemma 13 (Equality case)**.** Let $P = A_{i_r j_r} \cdots A_{i_1 j_1}$. If $f \in E_0$ and

$$\|Pf\| = \|f\|,$$

then $f_{i_s} = f_{j_s}$ for every averaged pair $\{i_s, j_s\}$.

Proof. Each $A_{i_s j_s}$ is an orthogonal projection, so norms can only decrease at each step. Equality at the end forces equality at every intermediate step. For the step A_{ij}, equality in the projection identity is exactly $f_i = f_j$ for the vector before that step. Since previous projections do not increase norm, the same conclusion propagates along the equality chain. □

Corollary 14 (Exact connected-block criterion)**.** Let H be the graph on $\{1, \ldots, N\}$ whose edges are the averaged pairs $\{i_s, j_s\}$. Then

$$\|A_{i_r j_r} \cdots A_{i_1 j_1}|_{E_0}\| < 1 \quad \Longleftrightarrow \quad H \text{ is connected.}$$

Proof. If H is disconnected, a nonzero mean-zero field constant on its components is fixed by every factor. Conversely, if equality held for some nonzero $f \in E_0$, the lemma would force f to be constant on every component of H. Connectedness would make it globally constant, hence zero, a contradiction. □

Proposition 15 (Quantitative connected-block contraction)**.** Let m pair averages have connected union graph, and put $a_N = 2(1 - \cos(\pi/N))$. Then

$$\|A_{i_m j_m} \cdots A_{i_1 j_1}|_{E_0}\|^2 \le 1 - \frac{a_N}{2(N-1)(1+\sqrt{m})^2}.$$

Consequently, if every consecutive block of m pulled-back effective pairs is connected, arbitrary settlements still give deterministic exponential consensus, with exponent at most

$$\frac{1}{2m} \log\left(1 - \frac{a_N}{2(N-1)(1+\sqrt{m})^2}\right) < 0.$$

Proof. Let x^s be the field after s averages, $d_s = x^{s-1}_{i_s} - x^{s-1}_{j_s}$, and

$$D = \mathcal{V}(x^0) - \mathcal{V}(x^m) = \frac{1}{2} \sum_s d_s^2.$$

The total displacement before any step is at most

$$\sum_s \left\|x^s - x^{s-1}\right\| = \sum_s \frac{|d_s|}{\sqrt{2}} \le \sqrt{mD}.$$

Choose one occurrence of each edge of a spanning tree of the union graph. At the time such an edge ij is averaged,

$$|x_i^0 - x_j^0| \le |d_s| + \sqrt{2}\left\|x^{s-1} - x^0\right\| \le \sqrt{2D}(1 + \sqrt{m}).$$

The spanning-tree Poincaré inequality and the path-minimality of algebraic connectivity give

$$a_N \mathcal{V}(x^0) \le \sum_{ij \in T} (x_i^0 - x_j^0)^2 \le 2(N-1)(1+\sqrt{m})^2 D.$$

Rearranging proves the one-block estimate. Multiplying the estimates over disjoint blocks proves the exponent bound. □

Thus the relevant connectivity condition is imposed on the endpoint pairs in one persistent frame, rather than on physical source edges written in successively changing frames.

8 Finite-N Exponential Consensus

The following theorem gives a robust fixed-N baseline. It says nothing sharp about the constant as N grows, and at some sizes the conclusion strengthens to finite-time exact consensus.

Theorem 16 (Fixed-N exponential consensus on the labelled lift)**.** Fix N. Let (X_t) be a finite irreducible Markov chain on labelled connected cubic multigraphs on the persistent vertex set $\{1, \dots, N\}$, and assume that every concrete Whitehead move available from a state has positive transition probability. Do not resettle between steps, and let

$$B_t = A_{i_t j_t}$$

be the field operator at time t, restricted to E_0. Then there exist $M \ge 1$, $q > 0$, and $\rho < 1$ such that almost surely

$$\limsup_{t \to \infty} \frac{1}{t} \log \|B_{t-1} \cdots B_0|_{E_0}\| \le \frac{q}{M} \log \rho < 0.$$

Consequently every initial field converges exponentially to its mean.

Proof. Choose one labelled connected cubic graph G^* and a spanning tree $T^* \subset G^*$. Since the labelled chain is finite and irreducible, from any state x there is a path of Whitehead moves leading to G^*. By finiteness, the length of such a path can be chosen at most D uniformly in x, after padding if necessary. Once at G^*, for each edge of T^* perform a chosen Whitehead move and immediately perform its inverse. This averages every tree edge and returns the labelled carrier to G^* after each pair. Let M be a common upper bound for the total length of this "success block".

Full support and finiteness imply that the conditional probability of realizing one of these success blocks in the next M steps is bounded below by some $q > 0$, uniformly over the current state.

For every success block, the product of its averaging projections includes all edges of the spanning tree T^*. By the connected-block corollary, its restriction to E_0 has norm strictly less than one. There are only finitely many success-block words of length at most M, so the maximum of their norms is some $\rho < 1$.

Now divide time into disjoint blocks of length M. Let Y_k be the indicator that block k is successful and put $p_k = \mathbb{E}[Y_k \mid \mathcal{F}_{kM}] \ge q$. The martingale $\sum_{k<r}(Y_k - p_k)$ has bounded increments, so its strong law [11, 12] gives

$$\liminf_{r \to \infty} \frac{1}{r} \sum_{k<r} Y_k \ge q \quad \text{almost surely.}$$

Successful blocks contract by at most ρ; all other blocks have norm at most one. Hence

$$\limsup_{r \to \infty} \frac{1}{rM} \log \|B_{rM-1} \cdots B_0|_{E_0}\| \le \frac{q}{M} \log \rho.$$

The leftover times between multiples of M contribute only norm-one factors, so the same limsup holds for all t. □

Remark 17. The theorem is deliberately stated on the persistent labelled lift. For an unlabelled implementation with intermediate settlements $P_{\sigma_t} A_{i_t j_t}$, orthogonality of each permutation alone does not preserve the connected-block argument: cumulative settlements conjugate the later averaging pairs. The same proof works if the averaged pairs are pulled back to one persistent frame and those effective pairs form a connected block. Without that condition, fixed-N consensus for an arbitrary settlement rule has not been proved here.

Theorem 18 (Finite-time consensus at $N = 2, 4, 8$)**.** Under the hypotheses of the persistent labelled theorem, the restriction of the field product to E_0 becomes exactly zero in finite time almost surely at $N = 2, 4, 8$. Thus its top exponent is $-\infty$ at these sizes.

Proof. A Whitehead move on a carrier edge e, immediately followed by its graph inverse, returns the labelled carrier and applies $A_e^2 = A_e$ to the field. It therefore embeds any finite edge-average word supported by one carrier into a returning Whitehead word.

At $N = 2$, one nonloop average is consensus. At $N = 4$, take K_4 and average the two perfect matchings

$$\{12, 34\}, \qquad \{13, 24\}.$$

The first round makes two equal-value pairs and the second merges them. At $N = 8$, take the cube Q_3 and average its three coordinate perfect matchings; the equal-value blocks double from sizes 1 to $2, 4, 8$. In each case the product is exactly $J = N^{-1}\mathbf{1}\mathbf{1}^T$.

The finite irreducible carrier chain visits the chosen carrier infinitely often. Full support gives a fixed positive probability of following the finite killing word at each visit, so the word occurs almost surely. Every later doubly stochastic factor fixes J. □

Proposition 19 (A three-value field that cannot reach exact consensus)**.** A finite product of pair averages and permutations can equal J only if N is a power of two. If N is not a power of two, then after t averaging steps

$$\|T_{t-1} \cdots T_0|_{E_0}\| = \|T_{t-1} \cdots T_0 - J\| \geq \frac{1}{N 2^t}.$$

Proof. Every entry of the product is dyadic with denominator dividing 2^t. Killing E_0 while fixing constants forces the full product to be J. Its entries $1/N$ are dyadic only when N is a power of two. Otherwise some entry differs from $1/N$ by at least $1/(N2^t)$, and the operator norm dominates every matrix entry. □

The proof also explains why the bridged bottleneck does not automatically destroy finite-N field relaxation. A successful block may pass through bridged graphs and may average bridge edges. That is fine: the block contracts because the list of averaged label-pairs contains a spanning tree, not because the intermediate base graphs avoid bottlenecks.

9 Bounds on How Fast the Field Reaches Consensus

For a product of settled averaging maps, define

$$\Lambda = \limsup_{t \to \infty} \frac{1}{t} \log \|T_{t-1} \cdots T_0|_{E_0}\| .$$

This number measures decay of the worst initial mean-zero field. If $\Lambda \le -r < 0$, then its disagreement is at most $\exp(-rt+o(t))$ times its initial size. A more negative value means faster consensus; $\Lambda = -\infty$ means that a finite product kills every mean-zero field exactly. This rate concerns the field values. It is not the mixing time of the graph-valued Markov chain.

The fixed-N theorem proves that Λ is negative but does not give a useful large-N scale. A random local average on an expander cubic graph suggests a bound of order $-1/N$.

There is first a bound that applies to every connected carrier trajectory; it does not assume expansion or stationarity.

Theorem 20 (A bound for every connected carrier trajectory)**.** For every connected cubic multigraph G on N vertices,

$$\lambda_2(L_G) \ge a_N := 2\big(1-\cos(\pi/N)\big).$$

Consequently every adapted trajectory of connected carriers under uniform-edge Rule A, with arbitrary settlement permutations, satisfies

$$\limsup_{t\to\infty} \frac{1}{t} \log \|T_{t-1}\cdots T_0|_{E_0}\| \le \frac{1}{2}\log\Big(1-\frac{a_N}{3N}\Big) = -\frac{\pi^2}{6N^3} + O(N^{-5}).$$

No irreducibility, stationary initial law, or expander frequency is required.

Proof. Delete loops and choose a spanning tree T of G, retaining one edge from a parallel family. Then $L_G \succeq L_T$. The path minimizes algebraic connectivity among trees on N vertices, and its gap is $2(1-\cos(\pi/N))$.

The conditional heat identity now gives

$$\mathbb{E}[\mathcal{V}(f_{t+1}) \mid \mathcal{F}_t] \le \Big(1-\frac{a_N}{3N}\Big)\,\mathcal{V}(f_t).$$

Settlement preserves variance. Dividing by the deterministic geometric factor produces a nonnegative supermartingale. Applying its almost-sure bound to a basis of E_0 gives the operator statement. □

The worst-carrier order cannot be improved from connectivity alone. Put a loop at each endpoint of a path and give consecutive path edges alternating multiplicities $1, 2, 1, 2, \dots, 1$. This is cubic, and the linear test field has the following ratio of Laplacian energy to squared norm:

$$\frac{18N-24}{N(N^2-1)} = O(N^{-2}).$$

Thus connectedness alone gives the N^{-3} bound, whereas typical stationary carriers give the stronger N^{-1} bound. They answer different questions.

Let (G_t, f_t) be a Rule A process in which, conditional on G_t, the averaged edge object is uniform among the $3N/2$ edge objects of G_t. Let

$$H_t = \{\lambda_2(L_{G_t}) \ge c_0\}, \qquad \alpha_N = \frac{c_0}{3N},$$

where $0 < c_0 < 3N$.

Theorem 21 (Frequent expanding carriers give $-c/N$ contraction)**.** Assume that for some $p \in [0,1]$,

$$\liminf_{t\to\infty} \frac{1}{t}\sum_{s=0}^{t-1} \mathbf{1}_{H_s} \ge p \qquad \text{almost surely.}$$

Then for every initial field f_0,

$$\limsup_{t\to\infty} \frac{1}{t} \log \mathcal{V}(f_t) \le p \log(1-\alpha_N) \qquad \text{almost surely.}$$

Equivalently, if f_0 is not already constant,

$$\limsup_{t\to\infty} \frac{1}{t} \log \left\| f_t - \bar{f}_0 \mathbf{1} \right\| \le \frac{p}{2} \log(1-\alpha_N) = -\frac{pc_0}{6N} + O(N^{-2}).$$

For fixed N, the same upper bound holds for the top operator exponent on E_0.

Proof. The conditional heat-step identity and the definition of ℓ_t imply

$$\mathbb{E}[\mathcal{V}(f_{t+1}) \mid \mathcal{F}_t] \le (1 - \alpha_N \mathbf{1}_{H_t}) \mathcal{V}(f_t).$$

Define

$$Q_t = \prod_{s=0}^{t-1} (1 - \alpha_N \mathbf{1}_{H_s}), \qquad Z_t = \frac{\mathcal{V}(f_t)}{Q_t}.$$

Since H_t is determined by G_t, the factor Q_{t+1}/Q_t is $\mathcal{F}_t$-measurable. Hence (Z_t) is a nonnegative supermartingale and therefore converges almost surely to a finite limit. Thus

$$\limsup_{t\to\infty} \frac{1}{t} \log \mathcal{V}(f_t) \le \limsup_{t\to\infty} \frac{1}{t} \log Q_t.$$

But

$$\frac{1}{t} \log Q_t = \frac{1}{t} \sum_{s=0}^{t-1} \mathbf{1}_{H_s} \log(1-\alpha_N),$$

and $\log(1-\alpha_N) < 0$. The assumed lower frequency of H_s gives the stated upper bound.

The norm bound is half the variance bound. For the operator norm on E_0, apply the vector bound to a fixed orthonormal basis of E_0 and intersect the finitely many almost-sure events. If every basis vector decays at rate at most r, then every vector decays at rate at most r up to a fixed dimension constant, which disappears after division by t. □

Theorem 22 (Spectral-average version)**.** Assume $N \ge 4$. For the same Rule A process, set

$$\ell_t = \lambda_2(L_{G_t}).$$

If

$$\liminf_{t\to\infty} \frac{1}{t} \sum_{s=0}^{t-1} \ell_s \ge \ell_* \qquad \text{almost surely,}$$

then

$$\limsup_{t\to\infty} \frac{1}{t} \log \mathcal{V}(f_t) \le \limsup_{t\to\infty} \frac{1}{t} \sum_{s=0}^{t-1} \log\left(1 - \frac{\ell_s}{3N}\right) \le -\frac{\ell_*}{3N},$$

and therefore

$$\limsup_{t\to\infty} \frac{1}{t} \log \left\| f_t - \bar{f}_0 \mathbf{1} \right\| \le -\frac{\ell_*}{6N}.$$

Proof. The conditional variance estimate holds with the actual current spectral gap:

$$\mathbb{E}[\mathcal{V}(f_{t+1}) \mid \mathcal{F}_t] \le \left(1 - \frac{\ell_t}{3N}\right) \mathcal{V}(f_t).$$

The factor is positive because the Laplacian eigenvalues of a cubic multigraph lie in $[0, 6]$ and $3N > 6$. (At $N = 2$, a zero factor means one-step consensus and can be handled separately.) Define

$$Q_t = \prod_{s=0}^{t-1} \left(1 - \frac{\ell_s}{3N}\right).$$

As before, $\mathcal{V}(f_t)/Q_t$ is a nonnegative supermartingale. This gives the first displayed bound. The second follows from

$$\log(1 - x) \le -x \qquad (0 \le x < 1).$$

□

This gives precise meaning to saying that the current graph is an expander on average. What is needed is not stationarity itself, but a pathwise lower bound on the time average of the cubic-graph spectral gap seen by the moving field.

Corollary 23 (Stationary law as one source of good frequency)**.** Suppose the graph component is a finite irreducible Markov chain with stationary law π_N. If

$$\pi_N\{G : \lambda_2(L_G) \ge c_0\} \ge 1 - \varepsilon_N,$$

then the theorem applies with $p = 1 - \varepsilon_N$, for every initial graph.

Proof. For a finite irreducible Markov chain, the ergodic theorem says that for every function φ on the state space,

$$\frac{1}{t} \sum_{s=0}^{t-1} \varphi(G_s) \to \sum_G \pi_N(G)\varphi(G) \quad \text{almost surely.}$$

Apply this to the indicator of the good set. □

For concrete Rule A, the preceding stationary law is not abstract: it is the cubic configuration model conditioned on connectedness. Spectral typicality can therefore be imported from the matching model itself rather than posed as a new property of an unknown stationary law.

This corollary is not the only route. Stationarity is convenient, but the Lyapunov theorem only needs a pathwise lower frequency of good graphs. A statement such as "the graph is an expander on average" is not by itself enough unless it implies such a frequency or a comparable supermartingale inequality.

The Jensen/log heuristic is consistent with the supermartingale proof. Whenever $\mathcal{V}(f_t) > 0$,

$$\mathbb{E}\left[\log \frac{\mathcal{V}(f_{t+1})}{\mathcal{V}(f_t)} \,\middle|\, \mathcal{F}_t\right] \le \log \mathbb{E}\left[\frac{\mathcal{V}(f_{t+1})}{\mathcal{V}(f_t)} \,\middle|\, \mathcal{F}_t\right].$$

On a good graph this is at most $\log(1 - c_0/(3N))$. The supermartingale argument is stronger because it converts conditional expected contraction into an almost-sure pathwise upper bound.

10 Exact One-Step Identities Beyond Variance

The configuration marginal gives a useful reset benchmark, but it must not be mistaken for the correlated trajectory law.

Proposition 24 (Independent configuration reset)**.** Let G be an unconditioned cubic configuration, let e be a uniform matched edge object, and let a deterministic mean-zero f be independent of G. Then exactly

$$\mathbb{E}\mathcal{V}(A_e f) = \left(1 - \frac{3}{3N-1}\right)\mathcal{V}(f),$$

or equivalently

$$\mathbb{E}L_G = \frac{9N}{3N-1}\left(I - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right).$$

If $\ell_N = \mathbb{E}[\#\text{ loops} \mid G \text{ connected}]$, the connected reset factor is

$$1 - \frac{3N - 2\ell_N}{3N(N-1)}.$$

Proof. Expose an oriented uniform stub. Conditional on its vertex being i, each $j \neq i$ is the partner vertex with probability $3/(3N-1)$. Hence

$$\mathbb{E}(f_i - f_j)^2 = \frac{3}{N(3N-1)}\sum_i \sum_{j\neq i}(f_i - f_j)^2 = \frac{6}{3N-1}\mathcal{V}(f),$$

and one average loses half of this. Under connected conditioning, permutation invariance makes the mean Laplacian scalar on E_0; its trace $3N - 2\ell_N$ determines the scalar. □

Fresh independent resampling would iterate this factor and suggest norm constant $1/2$ on the N^{-1} scale. The actual field records previously selected edges and is correlated with G_t, so the identity does not iterate along a Whitehead trajectory.

Theorem 25 (Resolution-independent target energy)**.** Suppose a move on $e = \{i, j\}$ produces either raw resolution G'. Then

$$(A_e f)^T L_{G'}(A_e f) = (A_e f)^T L_G (A_e f).$$

For $b_e = e_i - e_j$ (and $b_e = 0$ on a loop), put

$$\kappa_e = b_e^T L_G b_e, \qquad D_G = \sum_e \kappa_e b_e b_e^T, \qquad M = \frac{3N}{2}.$$

Uniform-edge Rule A therefore satisfies

$$\mathbb{E}[f'^T L_{G'} f' \mid G, f] = f^T\left(L_G - \frac{1}{M}L_G^2 + \frac{1}{4M}D_G\right)f.$$

For a simple cubic carrier, $D_G = 8L_G$.

Proof. After A_e, the values at i, j agree. A Whitehead resolution only changes which of these equal-valued endpoints owns each surrounding half-edge, so it does not change any contrast energy. Averaging $A_e L_G A_e$ over edge objects and using $\sum_e b_e b_e^T = L_G$ gives the matrix identity. In a simple cubic graph, $b_e^T L_G b_e = 3 + 3 + 2 = 8$. □

Thus, for a simple carrier and an L_G-eigenfield of eigenvalue λ, the expected carrier energy is multiplied by $1 + (2 - \lambda)/M$. When $\lambda < 2$, that energy can increase even while variance decreases; the next moment involves L_G^2 and exposes the carrier–field alignment problem.

Proposition 26 (Exact cut drift and cut count)**.** Let $|S| = k$, $b = |\partial_G S|$, and $f = \mathbf{1}_S - (k/N)\mathbf{1}$. Then

$$\frac{\mathbb{E}[\mathcal{V}(f') \mid G, f]}{\mathcal{V}(f)} = 1 - \frac{b}{3k(N-k)}.$$

In an unconditioned cubic configuration, the expected number of k-vertex sets with exactly b crossing edge objects is

$$\binom{N}{k} \frac{\binom{3k}{b}\binom{3(N-k)}{b} b!(3k-b-1)!!(3(N-k)-b-1)!!}{(3N-1)!!},$$

when $3k - b$ is even, and zero otherwise. For fixed B and $\varepsilon > 0$, the probability of a cut satisfying

$$\varepsilon N \le |S| \le N/2, \qquad |\partial S| \le B,$$

is at most $e^{-c_{\varepsilon,B} N}$ for all sufficiently large N; this persists after conditioning on connectedness.

Proof. Only a crossing edge changes the two-level field, losing $1/2$ unit of variance; divide the expected loss $b/(3N)$ by $k(N-k)/N$. For the count, choose the b crossing stubs on both sides, match them bijectively, and pair the remaining stubs internally. Stirling's formula shows that the logarithm of this probability is at most $-NH(k/N)/2 + o(N)$ uniformly on the displayed balanced range. □

11 Random Cubic Expansion and GoG Bottlenecks

For a simple random cubic graph, Friedman's theorem gives, with probability tending to one,

$$\lambda_2(L_G) \ge 3 - 2\sqrt{2} - \eta$$

for every fixed $\eta > 0$ [7]. More directly for present purposes, Bordenave proves the corresponding non-backtracking bound for a uniform perfect matching of the cubic half-edges [8]. The Ihara–Bass relation for regular multigraphs gives

$$\lambda_2(L_G) \ge 3 - 2\sqrt{2} - o(1)$$

with high probability in the cubic configuration model. Random cubic configurations are connected with high probability, so conditioning on connectedness does not change this conclusion. Thus the number

$$3 - 2\sqrt{2} \approx 0.1716$$

is the natural benchmark lower gap for a typical Rule-A carrier.

Corollary 27 (Stationary Rule-A field-contraction bound)**.** For any specified adapted settlement rule, define the upper top norm exponent by

$$\Lambda_N^A := \limsup_{t\to\infty} \frac{1}{t} \log \|T_{t-1} \cdots T_0|_{E_0}\| .$$

For the stationary Rule-A field process, and from every initial graph by ergodicity,

$$\Lambda_N^A \le -\frac{3 - 2\sqrt{2} - o(1)}{6N}$$

for the top norm exponent.

Proof. The configuration-model stationary identity and the preceding spectral theorem give good-graph frequency $1 - o(1)$. Apply the good-graph-frequency theorem with any $c_0 < 3 - 2\sqrt{2}$ and then let c_0 increase to the benchmark. □

Proposition 28 (Sharp GDPK bridge flow for Rule A)**.** Let B_N be the bridged classes, $B_0 = B_N^c$, and B_1 the classes with exactly one bridge. For class sets S, T, write

$$Q_A(S,T) = \sum_{\substack{G \in S \\ H \in T}} \pi_A(G) P_A(G,H),$$

and normalize a cut by its smaller stationary side:

$$\Phi_A(B_N) = \frac{Q_A(B_N, B_0)}{\min\{\pi_A(B_N), \pi_A(B_0)\}}.$$

Then exactly

$$Q_A(B_N, B_0) = \frac{2}{3N} \pi_A(B_1).$$

Moreover,

$$\pi_A(B_0) \to e^{-1}, \qquad \pi_A(B_1) \to e^{-1}, \qquad \pi_A(B_N) \to 1 - e^{-1},$$

and hence

$$\Phi_A(B_N) = \frac{2}{3N}(1 + o(1)).$$

In particular the Rule-A gap is $O(N^{-1})$ and the relaxation time is $\Omega(N)$.

Proof. A move can cross from the bridged to the bridgeless side only by acting on the unique bridge. Conversely, let $e = uv$ be the unique bridge and let A, C be the two components of $G - e$. Neither component has an internal bridge. In either resolution, one side edge at u and one side edge at v change endpoint, and the two connected sides are then joined by those two edges and e. Each of these three cross edges lies on a cycle using the other two and paths inside A, C; cycles internal to either side are unchanged or repaired through e. Thus both compressed resolutions are bridgeless. A class in B_1 has exactly two exiting slots, while a class with more than one bridge has none. Averaging under π_A proves the exact flow formula.

It remains to determine the masses. In the unconditioned configuration model, the loop count has mean $3N/(3N-1)$ and converges by factorial moments to Pois(1). If a bridge cuts off a smaller vertex set of size k, then $3k = 2|E(S)| + 1$, so k is odd. The expected number of k-sets with exactly one crossing matched pair is

$$A_{N,k} = \binom{N}{k} (3k)(3N - 3k) \frac{(3k-2)!!(3N-3k-2)!!}{(3N-1)!!}.$$

This is a first-moment upper bound because the two internally paired pieces need not be connected. The first term is

$$A_{N,3} = \frac{35}{18N} + O(N^{-2}).$$

Exact double-factorial ratios give $A_{N,k} = O(N^{-(k-1)/2})$ for each fixed odd k, and a uniform Stirling bound makes the middle range exponentially small. Consequently

$$\sum_{\substack{3 \le k \le N/2 \\ k \text{ odd}}} A_{N,k} = \frac{35}{18N} + O(N^{-2}) = O(N^{-1}).$$

Conditioning on connectedness changes these estimates by $1 + O(N^{-1})$. Indeed, the analogous expected count of an isolated even k-set is

$$D_{N,k} = \binom{N}{k} \frac{(3k-1)!!(3N-3k-1)!!}{(3N-1)!!};$$

its $k = 2$ term is $5/(18N) + O(N^{-2})$, and the same ratio and Stirling estimates give total disconnection probability $O(N^{-1})$.

A loop vertex has one remaining incident edge, which is a singleton-side bridge. Up to an event of probability $O(N^{-1})$, all bridges are of this form. Hence B_0 differs from $\{L = 0\}$ by probability $O(N^{-1})$ and B_1 differs from $\{L = 1\}$ by probability $O(N^{-1})$. The Poisson limit gives the three stationary-mass limits, and substitution in the exact flow formula gives the conductance asymptotic. □

Starting from stationarity conditioned on B_0 and testing total variation on B_0 gives, for fixed $\varepsilon < 1 - e^{-1}$,

$$t_{\mathrm{mix}}(\varepsilon) \geq \frac{3N}{2}(1 - e^{-1} - \varepsilon)(1 + o(1)).$$

by the standard bottleneck argument [10]. Thus $t_{\mathrm{mix}}(1/4) \geq 0.57318\ldots N(1+o(1))$. The stationary mean bridgeless and bridged run lengths are respectively

$$\frac{3N}{2}(1 + o(1)), \qquad \frac{3N}{2}(e-1)(1 + o(1)).$$

Conjecture 29 (Finite bridge identity)**.** For every even $N \geq 4$,

$$\Phi_A(B_N) = \frac{1}{3N/2 - 1}.$$

Equivalently, if C_0 and C_1 denote the total half-edge configuration weights of B_0 and B_1, respectively, then $(3N/2 - 1)C_1 = (3N/2)C_0$. This identity has been verified only for $N \leq 14$; the preceding proposition proves only its asymptotic consequence.

This is a genuine reversible Markov-conductance consequence. It is distinct from GDPK's directed outer-conductance, whose volume is cardinality times the constant outgoing slot count.

Proposition 30 (Increasingly many vanishing Rule-A eigenvalues)**.** For every fixed k, the kth eigenvalue of the reversible normalized Rule-A Laplacian $I - P_A$ satisfies

$$\lambda_k(I - P_A) \leq \frac{C_k}{N}$$

for all sufficiently large even N.

Proof. Choose $k - 1$ distinct fixed cycle lengths. Under the cubic configuration model, the corresponding cycle-count variables converge jointly to independent Poisson variables. Hence the centered indicators that a carrier contains a cycle of each chosen length have norms bounded away from zero and an asymptotically nonsingular Gram matrix.

If a carrier contains a chosen j-cycle, a one-step move which destroys the last such cycle must be performed on or within bounded distance of that cycle. Only $O_j(1)$ of the $3N$ compressed slots can do this. The Rule-A Dirichlet energy of its centered indicator is therefore $O_j(N^{-1})$. Cauchy–Schwarz gives the same order for cross energies. The min–max principle on the span of these indicators and the constant function gives the result. □

This is the normalized multiplicity-chain analogue of Li's bounded-eigenvalue construction. It concerns the Markov operator actually sampled by local Rule A, rather than either simple-support Laplacian.

The key distinction is:

$$\text{GoG bottleneck} \quad \neq \quad \text{bad field-averaging graph.}$$

Grave de Peralta–Kolpakov's bridged set is a bottleneck in Γ_N because few Whitehead moves leave the set of bridged graphs. But the field relaxation at a time t is controlled by $\lambda_2(L_{G_t})$ or by the weighted analogue coming from the edge-selection rule. A graph with a microscopic bridge may still have a spectral gap bounded below by a constant. A balanced bridge is bad for field averaging, because it gives a cut of size one separating two linear-size pieces; a function constant on the two sides then gives $\lambda_2(L_G) = O(1/N)$.

More explicitly, if a bridge separates a set S of k vertices from its complement, the test function which is constant on S and on S^c gives

$$\lambda_2(L_G) \le \frac{N}{k(N-k)}.$$

This is $O(1/N)$ for balanced bridges and $O(1)$ for microscopic bridges. Thus bridged graphs are not uniformly bad for field relaxation; balanced sparse cuts are.

The converse direction is governed by ordinary spectral graph theory. Define the edge isoperimetric constant of the current cubic graph by

$$h(G) = \min_{0<|S|\le N/2} \frac{|\partial_G S|}{|S|},$$

where $\partial_G S$ is the set of edge objects crossing from S to S^c. For graphs of maximum degree 3, the combinatorial Cheeger inequalities give the explicit bounds [5]

$$\frac{h(G)^2}{6} \le \lambda_2(L_G) \le 2h(G).$$

Thus a large sparse cut supplies a slowly contracting field direction, not a statement that every current field contracts slowly. The actual one-step loss is governed by $f^T L_G f/\mathcal{V}(f)$; a field nearly orthogonal to an eigenvector with small Laplacian eigenvalue may contract quickly. Microscopic bridges can coexist with $h(G) \asymp 1$ when all cuts separating a linear-size set have linear boundary, whereas balanced bridges force $h(G) = O(1/N)$. This is the precise interface between bridge geometry and the Lyapunov estimate.

The Rule-A stationary probability of a balanced cut of any fixed width consequently tends to zero. The bridged GoG bottleneck remains important for the mixing of the base chain, but it does not bias its equilibrium carrier marginal away from the configuration model.

Conjecture 31 (Expected $1/N$ contraction scale without a killing word)**.** For a precisely specified Rule-A cocycle with no finite consensus word, or along sizes for which such a word is impossible, the finite top exponent satisfies

$$N\Lambda_N \to -\theta$$

for some $\theta > 0$. The value may differ from the independent reset constant because the evolving field and carrier are correlated. Without the stated qualification, the conjecture is false as a finite-valued assertion at $N = 2, 4, 8$, and possibly at further powers of two.

12 Results for the three graphs of moves

The preceding field analysis uses the multiplicity-first Rule-A clock. The three base graphs also admit sharper, mutually incompatible conclusions.

12.1 The GDPK multiplicity chain

Theorem 32 (Spectral interval, Perron root, and history entropy)**.** Let $M_N = (m_{GH})$ be the $3N$-slot adjacency matrix and $P_A = M_N/(3N)$. Then

$$\mathrm{Spec}(P_A) \subset [-1/2, 1], \qquad \rho(M_N) = 3N.$$

In particular M_N is real-diagonalizable, and from every source class the number of concrete length-t slot histories is exactly $(3N)^t$. Furthermore the stationary entropy rate of the class-valued chain is

$$h_N = \log(3N) - \Delta_N, \qquad \Delta_N = \frac{1}{3N} \sum_G \pi_A(G) \sum_H m_{GH} \log m_{GH}.$$

If $C_1(G) = \sum_H \binom{m_{GH}}{2}$, then

$$0 \le \Delta_N \le \frac{2}{3N} \mathbb{E}_{\pi_A} C_1(G).$$

Proof. Detailed balance symmetrizes P_A. Upstairs, freeze a central nonloop edge and the four surrounding labelled stubs. Their six two–two assignments form the Johnson graph $J(4,2)$, whose normalized adjacency spectrum is $\{1, 0, -1/2\}$. A loop gives the identity. Averaging these local kernels and restricting to class-invariant functions proves the spectral interval. The row sum of M_N is exactly $3N$, proving the Perron and history claims. The entropy formula follows by expanding the entropy of the row probabilities $m_{GH}/(3N)$; Δ_N is precisely the conditional entropy of the chosen slot given source and target classes. Finally, $m \log m \le m(m-1) = 2\binom{m}{2}$ for every positive integer m, which proves the collision bound. □

Theorem 33 (Exact loop-count Dirichlet form)**.** For $N \ge 4$, let $L(G)$ be the number of carrier loops and $\mu_N(k) = \pi_A\{L = k\}$. Then for every function f on the nonnegative integers,

$$\mathcal{E}_A(f(L), f(L)) = \frac{2}{3N} \sum_{k \ge 1} k \mu_N(k) \big(f(k) - f(k-1)\big)^2.$$

Consequently, for every fixed $r \ge 1$,

$$\lambda_{r+1}(I - P_A) \le \frac{2r}{3N} + o(N^{-1}).$$

With the convention $\alpha_{\mathrm{LS}} = \inf \mathcal{E}(F,F)/\operatorname{Ent}_{\pi_A}(F^2)$, also $\alpha_{\mathrm{LS}} \le 2/(3N) + o(N^{-1})$.

Proof. A state with k loops has exactly $2k$ compressed slots that destroy one loop. No move changes loop count by more than one when $N \ge 4$. Detailed balance supplies the reverse aggregate flow and gives the displayed form. Under π_A, L tends to Pois(1). Applying min–max to the first r Charlier polynomials, whose limiting immigration–death ratios of Dirichlet energy to variance are $1, \dots, r$, proves the eigenvalue bound. The indicator of $L = 0$ proves the log-Sobolev upper bound. □

The projected process $L(G_t)$ need not itself be Markov. The theorem is an exact stationary Dirichlet-form identity and an upper bound; it does not say that the smallest eigenvalues converge to those of a one-dimensional birth–death chain.

12.2 The fixed-labelled simple graph

Write Ω_N^{lab} for connected adjacency multigraphs on the fixed set $[N]$, $d(G)$ for Li's simple degree, and $\delta(G) = 6N - d(G)$. Three natural walks have different stationary laws:

walk	stationary law
simple neighbour	$d(G)/\sum_H d(H)$
uniform clock $I - L(\Gamma_{\mathrm{lab}})/(6N)$	uniform on Ω_N^{lab}
four-slot multiplicity	$c(G)/\sum_H c(H)$.

Denote the uniform law by U_N and the four-slot stationary law by π_{slot}.

For connected $N > 2$, put $L(G)$ = number of loops and $D(G)$ = number of doubled vertex pairs. Triple edges cannot occur, and exactly

$$c(G) = \frac{6^N}{2^{L(G)+D(G)}}.$$

Under the slot law, (L, D) tends to two independent Pois(1) variables; exponential tilting shows that under the uniform adjacency law it tends to two independent Pois(2) variables. Hence

$$\|\pi_{\mathrm{slot}} - U_N\|_{\mathrm{TV}} \longrightarrow \big\|\mathrm{Pois}(1)^{\otimes 2} - \mathrm{Pois}(2)^{\otimes 2}\big\|_{\mathrm{TV}} = 0.4385731106\ldots.$$

Li's good-edge estimate gives the stronger statement $\mathbb{E}_{U_N}\delta = O(1)$, and therefore the simple-neighbour stationary law is $O(N^{-1})$ in total variation from U_N. Thus the two natural labelled walks remain order-one apart even though their rows are typically $O(N^{-1})$ apart.

More precisely, let $P_U = I - L(\Gamma_{\mathrm{lab}})/(6N)$, let P_{simp} be the simple-neighbour kernel, and let $K(G, H)$ be the multiplicity among the $6N$ labelled slots. Put $P_{\mathrm{slot}}(G, H) = K(G, H)/(6N)$. With $k_0(G) = K(G, G)$,

$$\delta(G) = k_0(G) + \sum_{H \neq G : K(G,H) > 0} (K(G, H) - 1),$$

and direct summation gives

$$\|P_{\mathrm{simp}}(G, \cdot) - P_U(G, \cdot)\|_{\mathrm{TV}} = \frac{\delta(G)}{6N}, \qquad \|P_{\mathrm{slot}}(G, \cdot) - P_U(G, \cdot)\|_{\mathrm{TV}} \leq \frac{\delta(G)}{6N}.$$

These local estimates do not imply close stationary laws because the relaxation scale is itself of order N.

Theorem 34 (Many bounded labelled Laplacian eigenvalues)**.** For

$$W_N = \left\{ (a_{ij}) : a_{ij} = a_{ji},\ a_{ii} = 0,\ \sum_{j \neq i} a_{ij} = 0 \right\}, \qquad \dim W_N = \frac{N(N-3)}{2},$$

put $F_a(G) = \sum_{i<j} a_{ij} m_{ij}(G)$. There is an absolute C such that, for all sufficiently large even N,

$$\lambda_{N(N-3)/2+1}\big(L(\Gamma_{\mathrm{lab}}(N))\big) \leq C.$$

The same number of nonconstant modes have eigenvalue $O(N^{-1})$ for the uniform-clock and normalized simple-neighbour operators.

Proof. The N row-sum equations have full rank, which gives the displayed dimension, and W_N is the irreducible $[N-2,2]$ representation of S_N. Both the uniform covariance of F_a and its simple-edge Dirichlet form are S_N-invariant, so Schur's lemma makes each a scalar form on W_N.

Let $x(G) = (m_{ij}(G))_{i<j}$. The trace of the covariance form on W_N is $\mathbb{E}_{U_N} \| \operatorname{Proj}_{W_N} x(G)\|^2$. Since $\sum_{i<j} m_{ij}^2 = 3N/2 + O(L+D)$ in expectation, while the trivial and standard projections have bounded expected squared norm, division by $\dim W_N$ gives

$$\mathbb{E}_{U_N} F_a^2 = \left(\frac{3}{N} + O(N^{-2})\right) \|a\|^2 .$$

Each raw move changes only four pair coordinates. Summing the resulting squared increments over slots and taking the trace on W_N gives simple-edge Dirichlet energy $O(N^{-1})\,\|a\|^2$; invariance again makes the same bound uniform over $a \in W_N$. Its quotient by the covariance is $O(1)$, while the uniform-clock normalization contributes an additional factor $(6N)^{-1}$.

For the normalized simple walk the denominator is the invariant form $\mathbb{E}_{U_N}[d(G)F_a(G)^2]$. Its trace is $\Theta(N^2)$ because $d = 6N - \delta$, $\mathbb{E}_{U_N}\delta = O(1)$, and $\| \operatorname{Proj}_{W_N} x\|^2 = O(N)$ deterministically. Hence this form is $\Theta(1)\,\|a\|^2$ and the normalized quotient is $O(N^{-1})$. Adding the constant function and applying min–max proves all three assertions. □

Theorem 35 (Fixed-radius labelled balls)**.** Let $G \sim U_N$. For every fixed R,

$$|B_R^{\mathrm{lab}}(G)| = \frac{(6N)^R}{R!} + O_{\mathbb{P},R}(N^{R-1}), \qquad \mathbb{E}_{U_N}|B_R^{\mathrm{lab}}(G)| = \frac{(6N)^R}{R!} + O_R(N^{R-1}).$$

The error estimate is not asserted uniformly for growing R.

Proof. Call a raw move R-good when the radius-$(2R+2)$ carrier neighbourhood of its central edge has no loop or parallel edge and is a tree rooted at that edge. A word is separated if all of these neighbourhoods are vertex-disjoint. Standard fixed-subgraph counting in the cubic configuration model [6], followed by the exact loop/double-edge exponential tilt to U_N, preserves these fixed-radius moment bounds and shows that the expected number of non-good raw slots is $O_R(1)$. Once fewer than R separated supports have been chosen, only $O_R(1)$ of the $6N$ next raw slots meet one of them. Hence the expected number of ordered separated good words is

$$(6N)^R + O_R(N^{R-1}),$$

and the same estimate holds in probability.

The rewrites in a separated word commute. In the symmetric difference of the source and endpoint edge sets, the affected neighbourhoods are distinct components; within each, the unchanged central edge is the unique diagonal. Thus the source–endpoint pair recovers the R selected moves and their resolutions. Exactly the $R!$ orders of those commuting moves have the same endpoint. Words with repeated or intersecting supports, or with a non-good slot, have expected count $O_R(N^{R-1})$, because one free $O(N)$ choice is lost. Words of length less than R have the same lower order. Dividing the good ordered count by $R!$ proves the expectation estimate, and Markov's inequality applied to the exceptional-word count gives the in-probability form. □

Moreover

$$|\Omega_N^{\mathrm{lab}}| = (e^2 + o(1))\frac{(3N-1)!!}{6^N},$$

so maximum degree $6N$ implies $\operatorname{diam}\Gamma_{\mathrm{lab}}(N) \geq (3/2 - o(1))N$, and the same lower bound holds for the distance between two uniform labelled states with high probability.

Proposition 36 (Exact orbit quotient)**.** For an unlabelled class α, put $s_\alpha = N!/|\operatorname{Aut}_v(\alpha)|$, and let $b_{\alpha\beta}$ be the number of labelled simple neighbours of one representative of α lying in orbit β. Then

$$s_\alpha b_{\alpha\beta} = s_\beta b_{\beta\alpha}.$$

The orbit partition is strongly lumpable for the labelled simple-neighbour walk, with transition $b_{\alpha\beta}/d_\alpha$ and stationary mass proportional to $s_\alpha d_\alpha$. The uniform-clock walk lumps with off-diagonal and diagonal transitions

$$\overline{P}_U(\alpha,\beta) = \frac{b_{\alpha\beta}}{6N} \quad (\beta \neq \alpha), \qquad \overline{P}_U(\alpha,\alpha) = \frac{6N - d_\alpha + b_{\alpha\alpha}}{6N},$$

and stationary mass proportional to s_α. The unlabelled simple support records only whether $b_{\alpha\beta} > 0$ and is not, in general, either Markov quotient.

Proof. The symmetric group acts transitively on every orbit and by automorphisms of the labelled simple graph, so $b_{\alpha\beta}$ is representative-independent. Double-counting labelled edges between the two orbits gives reciprocity. Summing its rows gives the stated lumped transition and stationary laws. Discarding the values of $b_{\alpha\beta}$ retains reachability but not these probabilities. □

12.3 The unlabelled simple support

Reciprocity makes the support undirected. Let L_{supp} be its combinatorial Laplacian and let $q(G)$ be the number of distinct nonself neighbours of G. Because $q(G) \leq 3N$, the fixed-clock kernel

$$P_B = I - \frac{1}{3N} L_{\mathrm{supp}}$$

is stochastic. It is symmetric, so its stationary law is uniform on the unlabelled classes. Its eigenvalues are related to those of the unnormalized support Laplacian without any estimate of $q(G)$:

$$\lambda_k(I - P_B) = \frac{\lambda_k(L_{\mathrm{supp}})}{3N}.$$

Conditioning this fixed-clock chain on making a nonself move gives the jump chain

$$J_B(G,H) = \frac{1}{q(G)} \quad \text{when } H \neq G \text{ and } m_{GH} > 0, \qquad \pi_{J_B}(G) = \frac{q(G)}{\sum_K q(K)}.$$

The jump chain traverses the same paths but deletes the state-dependent holding times. Its spectrum and mixing time need not agree with those of P_B.

The repeated-target count remains useful for comparing the jump chain with Rule A. Put

$$M_+(G) = \sum_{H \neq G} m_{GH}, \qquad C_1^+(G) = \sum_{H \neq G} \binom{m_{GH}}{2}.$$

Here $C_1^+(G)$ is the number of unordered pairs of distinct nonself Rule-A slots that have the same target class. If Rule A is conditioned on making a nonself move, its row differs from the Rule-B jump row by exactly

$$\chi^2(p_A^+ \| J_B(G,\cdot)) = \frac{q(G)(M_+(G) + 2C_1^+(G))}{M_+(G)^2} - 1.$$

This equation says concretely what is lost when repeated Rule-A targets are collapsed: the discrepancy is determined by the number of pairs of slots that collide at the same target.

If $C_1(G) = \sum_H \binom{m_{GH}}{2}$ includes the self target, Cauchy–Schwarz also gives the exact deterministic lower bound

$$q(G) \geq \frac{(3N)^2}{3N + 2C_1(G)} - 1.$$

Indeed, at most $q(G) + 1$ targets, including the self target, have positive multiplicity, while

$$\sum_H m_{GH}^2 = 3N + 2C_1(G).$$

For a set X of unlabelled classes, let b_X be the number of undirected support edges with one endpoint in X and the other in X^c. The centered membership function of X has the following exact quotient under the fixed-clock Rule B:

$$R_B(X) = \frac{b_X}{3N}\left(\frac{1}{|X|} + \frac{1}{|X^c|}\right).$$

There is no degree in this formula. In comparison, the jump-chain quotient contains the degree volumes $V_X = \sum_{G \in X} q(G)$:

$$R_{J_B}(X) = b_X\left(\frac{1}{V_X} + \frac{1}{V_{X^c}}\right).$$

Proposition 37 (What short-cycle cuts imply for fixed-clock Rule B)**.** Let X be a class property determined by the presence of a fixed short cycle. Assume both X and X^c contain at least a fixed positive proportion of all unlabelled classes as $N \to \infty$. Then

$$\lambda_2(I - P_B) = O(N^{-1}).$$

More generally, fix k. Suppose $k - 1$ such class properties have centered membership functions whose class-uniform Gram matrix stays nonsingular. Then

$$\lambda_k(I - P_B) = O_k(N^{-1}).$$

These conclusions require the stated class-count assumptions, but no estimate of the support degrees $q(G)$.

Proof. Li's local argument gives $b_X \leq C_X|X|$: only a bounded number of moves near a fixed short cycle can destroy the last cycle of that length. The positive class proportions and the displayed formula for $R_B(X)$ therefore give $R_B(X) = O_X(N^{-1})$. The min–max principle applied to the constant function and the centered membership function gives the first conclusion.

For finitely many class properties, the same boundary argument bounds every diagonal and cross Dirichlet form by $O_k(N^{-1})$ after class-uniform normalization. The assumed lower bound on the Gram matrix prevents a nonconstant linear combination from having vanishing variance. Applying min–max to their span gives the second conclusion. □

The unresolved input is now explicit: one must count unlabelled classes with and without the chosen short cycles. The previous formulation also required linear lower bounds on the mean support degrees inside every class set. The fixed clock removes that unnecessary requirement.

Proposition 38 (When a Rule-B step has zero field contraction)**.** Call a carrier edge active when at least one resolution has a nonself target. Assume Rule B gives positive probability to every support neighbour and every concrete lift of that transition. Then

$$\ker L_G^B = \{f : f \text{ is constant on every component of the active-edge carrier}\}.$$

The weighted gap is zero exactly when that carrier is disconnected, with mean-zero kernel dimension one less than its component count. This is a statement about the current step. A later rewrite can change the active components and restore contraction.

Proof. Under the full-support lift rule, an edge has positive averaging weight exactly when it is active. Since

$$f^T L_G^B f = \sum_e p_G^B(e)(f_i - f_j)^2,$$

the quadratic form vanishes precisely when $f_i = f_j$ along every active edge, which is equivalent to constancy on each active component. Intersecting this kernel with $\mathbf{1}^\perp$ removes one dimension. □

Finally, multiplicities and self-slots do not change reachability. Hence the simple support has exactly the GDPK support metric and inherits its $\Theta(N \log N)$ diameter order [9], even though its path counts and random-walk times differ.

13 Complete finite enumeration

The finite statements in this section are computational evidence, not asymptotic theorems. Cubic multigraphs were encoded by coloured incidence graphs and canonically labelled with nauty/Traces [4]. Enumeration is complete through $N = 14$: the class counts agree with A005967, and every source has exactly $3N$ compressed resolution slots.

N	unlabelled classes	compressed slots
2	2	12
4	5	60
6	17	306
8	71	1 704
10	388	11 640
12	2 592	93 312
14	21 096	886 032

13.1 Carrier and base-chain spectra

The minimum carrier gap over all classes is compared below with the connected-graph path bound $a_N = 2(1 - \cos(\pi/N))$.

N	$\min_G \lambda_2(L_G)$	a_N	$\min_G \lambda_2(L_G)/a_N$
2	2.000000	2.000000	1.000
4	0.763932	0.585786	1.304
6	0.354249	0.267949	1.322
8	0.202067	0.152241	1.327
10	0.130145	0.097887	1.330
12	0.090687	0.068148	1.331
14	0.066764	0.050144	1.331

At every enumerated $N \geq 4$, the minimizer is the alternating single/double-edge path with endpoint loops used in the proof of the $O(N^{-2})$ extremal upper bound.

The first six eigenvalues of the Rule-A kernel, the degree-normalized Rule-B jump chain, and the unnormalized support Laplacian are:

N	operator	1	2	3	4	5	6
8	$I - P_A$	.000	.046	.097	.124	.161	.194
	$I - J_B$	.000	.074	.163	.221	.275	.314
	L_{supp}	.000	.307	.617	.935	1.048	1.153
10	$I - P_A$	.000	.031	.063	.082	.087	.107
	$I - J_B$	.000	.046	.099	.135	.163	.187
	L_{supp}	.000	.307	.504	.757	.870	.999
12	$I - P_A$	.000	.023	.045	.060	.065	.077
	$I - J_B$	.000	.030	.065	.092	.105	.117
	L_{supp}	.000	.335	.682	.879	1.057	1.135
14	$I - P_A$	.000	.018	.034	.046	.050	.056
	$I - J_B$	.000	.021	.045	.066	.070	.080
	L_{supp}	.000	.332	.706	.788	.862	1.116

The canonical fixed-clock Rule-B eigenvalues are obtained by dividing the last row in each block by $3N$. Its second eigenvalues at $N = 8, 10, 12, 14$ are approximately $.0128, .0102, .0093, .0079$. These finite values do not prove a matching lower bound for the support Laplacian.

13.2 Bridge cut and zero one-step Rule-B contraction

The bridge-cut conductances are

N	support boundary edges	$\Phi_A(B_N)$	$\Phi_{J_B}(B_N)$
8	15	.0909	.2113
10	82	.0714	.1673
12	537	.0588	.1221
14	4 425	.0500	.0922

The Rule-A column agrees exactly with $1/(3N/2-1)$ for every even $N \leq 14$. This finite identity is a conjecture beyond the enumerated range; only its asymptotic consequence $\Phi_A(B_N) = 2/(3N)(1+o(1))$ is proved.

The full-support Rule-B field lift has a disconnected active-edge carrier on the following classes:

N	zero-gap classes	π_B-mass	π_A-mass	π_{J_B}-mass
8	17	.239	.139	.215
10	89	.229	.101	.195
12	548	.211	.0757	.174
14	4 102	.194	.0616	.1565

Because fixed-clock Rule B has uniform stationary law, its mass column is the class fraction. Every zero-gap class in the enumeration is bridged. The active-edge theorem makes the zero one-step gaps exact; the table does not imply a zero long-time field exponent.

13.3 Fixed-labelled graph

The labelled simple graph was materialized completely for $N \leq 6$.

N	states	simple edges	$\mathbb{E}_U d$	$\min d / \max d$	$\lambda_2(L_{\text{lab}})$
2	2	1	1.0000	1/1	2.000000
4	35	99	5.6571	5/6	2.052917
6	3 550	23 400	13.1831	9/24	1.646846

For larger N, exact orbit weighting gives

N	$\mathbb{E}_U d$	$6N$
8	22.232	48
10	32.105	60
12	42.501	72
14	53.266	84

The sizes shown are not yet in the asymptotic $6N - O(1)$ regime, but the ratios increase monotonically over this range. Exact rational arithmetic also verifies the K_4 and Q_3 consensus products, and direct evaluation of the resolution-independent energy identity on sampled transitions agrees to machine precision.

Reproducibility. The script `gog\_generate.py` constructs the canonical class and transition tables. The reported multiplicity/support spectra and bridge statistics are produced by `gog\_edge\_model\_experiments.py`; `gog\_support\_feature\_analysis.py` evaluates short-cycle and Rule-B diagnostics, and `gog\_ball\_growth.py` samples support balls. Labelled-degree and automorphism checks are implemented in `gog\_degree\_diagnostics.py`. Exact commands, conventions, and numerical limitations are recorded in `README.md`.

14 Lean Verification Boundary

A self-contained Lean 4 project in `lean/` kernel-checks the exact finite algebra that is actually formalized. It uses exact rational arithmetic and the standard library only; the project source contains no `sorry`, `admit`, or project-declared `axiom`. Representative `#print axioms` checks report only Lean's foundational principles `propext`, `Classical.choice`, and `Quot.sound`. The checked declarations include:

- the coordinate definition of pair averaging, symmetry, affine and centering covariance, preservation of the selected-pair sum, idempotence, and the exact squared-norm/variance drop $(x_i - x_j)^2/2$ with an arbitrary unchanged remainder;
- commutation of averages on disjoint pairs;
- identity, composition, and inverse laws for pullback by an explicit coordinate permutation, including exact conjugacy between relabelling and pair averaging;
- preservation of total mass, commutation and idempotence of the block schedules, and exact consensus for the two transverse K_4 matchings and the three coordinate matchings of Q_3;
- a common-denominator proof that no finite pair-average schedule sends the three-state datum $(0, 0, 1)$ to exact consensus;

- concrete 2×2 identities $A^2 = A$, $A\mathbf{1} = \mathbf{1}$, and agreement between matrix and coordinate pair averaging.

From the `lean/` directory, `lake build` checks the project. The declarations are in `lean/GogVerification/Averaging.lean`.

This is deliberately not advertised as a formalization of the whole paper. Lean does not presently check the Whitehead graph or its quotient, construct settlements, prove their coherence, or address stabilizer holonomy. Nor does it check the configuration counts, the representation-theoretic argument, random-graph inputs, asymptotics, or Lyapunov and mixing theorems. The dyadic result is a concrete three-state obstruction rather than a classification of finite consensus protocols. The formal layer certifies only the finite averaging identities on which several of those arguments rely.

15 Open Problems

Question 39. After fixing a settlement cocycle and excluding sizes with finite killing words, is the top Lyapunov exponent asymptotic to $-\theta/N$? If yes, is θ determined only by typical cubic expansion, or does it remember carrier–field alignment and projection noncommutativity?

Question 40. Can the square-defect curvature be related to GoG eigenvectors with small Laplacian eigenvalues? The plausible mechanism is that local motifs responsible for bottleneck eigenvectors also create high curvature density for incident Whitehead moves, but this is a testable statement rather than a theorem.

Question 41. Does the finite bridge identity $(3N/2 - 1)C_1 = (3N/2)C_0$ hold for all even N, and can one prove a matching c/N Poincaré lower bound for the labelled local Johnson-update lift?

Question 42. Under the uniform law on unlabelled classes, does every fixed short-cycle property used above occupy a positive limiting fraction of the classes, and do finite families of their centered membership functions have a nonsingular limiting Gram matrix? These statements would make the fixed-clock Rule-B $O(N^{-1})$ eigenvalue bounds unconditional without any support-degree estimate. In the other direction, can one prove $\lambda_2(L_{\mathrm{supp}}) \geq c > 0$, which together with $I - P_B = L_{\mathrm{supp}}/(3N)$ would give a matching $c/(3N)$ lower bound?

Question 43. Is there a useful Hodge-like theory for the covariant difference D on the field bundle over Γ_N? Such a theory must first specify edge-to-two-cell differentials for a noninvertible semigroup cocycle and treat automorphism groups as finite gauge stabilizers.

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